A

Song sparrow
Swamp sparrow
Savannah sparrow
White-throated sparrow
Red-winged blackbird
Common yellowthroat
Chaffinch
House finch
House sparrow
Zebra finch
European robin
Swainson's thrush
White-throated dipper
Common myna
Eurasian blackcap
Garden warbler
Silvereye
Barn swallow
Black-capped chickadee
New Caledonian crow
Blue jay
Paradise crow
Rifleman
Blue-and-yellow macaw
Swift parrot
Gyrfalcon
Lanner falcon
Peregrine falcon
Lesser kestrel
Red-legged seriema
Downy woodpecker
Carmine bee-eater
Ground hornbill
Common buzzard
White-tailed eagle
Harpy eagle
Golden eagle
Bearded vulture
King vulture
Plumbeous ibis
Eurasian spoonbill
Great cormorant
Northern gannet
Maguari stork
Humboldt penguin
Cory's shearwater
Red-throated loon
Red-billed tropicbird
Hoatzin
Herring gull
Lesser black-backed gull
Black-legged kittiwake
Common tern
Eurasian curlew
European golden plover
Northern lapwing
Takahe
Common moorhen
Common crane
Turtle dove
Collared dove
Rock dove
Band-tailed pigeon
Nicobar pigeon
Yellow-throated sandgrouse
American flamingo
MacQueen's bustard
Common cuckoo
Whiskered treeswift
Common swift
Long-billed hermit
Tawny frogmouth
European nightjar
Tufted duck
Greater scaup
Common pochard
Red-crested pochard
Mallard
Northern pintail
Common goldeneye
Long-tailed duck
Tundra swan
Mute swan
Greylag goose
Canada goose
Rock ptarmigan
Western capercaillie
Chicken
Crested tinamou
Darwin's rhea
Emu
Brown kiwi

-0.5 0.0 1.0 2.0
Technical innovation rate

B

DAR enrichment
Wilcoxon, BH FDR across 46 tests

Negative Rho

OL ***
AST ***
MGL
VASC
OPC **
EXC_GLU-8 **
IOL
INH_VIP
EXC_GLU-5
EXC_GLU-3
EXC_GLU-7
INH_1
INH_SST
EXC_GLU-6
EXC_GLU-1
EXC_GLU-2
INH_LAMP5
INH_2
EXC_GLU-4
INH_STR
D2MSN
D1MSN
INH_PVALB

Positive Rho

OL
AST
MGL
VASC
OPC
EXC_GLU-8
IOL
INH_VIP
EXC_GLU-5
EXC_GLU-3
EXC_GLU-7
INH_1
INH_SST
EXC_GLU-6
EXC_GLU-1
EXC_GLU-2
INH_LAMP5
INH_2
EXC_GLU-4
INH_STR
D2MSN
D1MSN
INH_PVALB

-0.06 -0.03 0.00 0.03
Effect size

FDR < 0.05 n.s.

C

LOC107055098 (-48 bp) ρ = 0.39, *P* = 0.011
TFAP2B (+249 bp) ρ = 0.3, *P* = 0.004
XCL1 (-277 bp) ρ = 0.38, *P* = 0.002
FGF10 (-353 bp) ρ = 0.33, *P* = 0.007
TSKU (+658 bp) ρ = 0.39, *P* = 0.003
PCK1 (-276 bp) ρ = −0.2, *P* = 0.008
LGI1 (-338 bp) ρ = −0.22, *P* = 0.033
GYPC (+590 bp) ρ = −0.22, *P* = 0.004
GJC3 (-690 bp) ρ = −0.25, *P* = 0.003
MIR1705 (+960 bp) ρ = −0.33, *P* = 6.8 × 10^{-6}

Technical innovation rate
Relative evolutionary rate (RER)